

# Yongming Liang 梁永明



## ICRR Fellow

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## Research interests

Galaxy evolution; the galaxy–IGM connection; quasars and supermassive black holes; protoclusters and large-scale structure; high-redshift galaxies and cosmic reionization.

## Appointments

- **2025-10–present · ICRR Fellow**  
ICRR, The University of Tokyo
- **2024-10–2025-09 · Project Researcher**  
National Astronomical Observatory of Japan
- **2022-10–2024-09 · Project Researcher**  
ICRR, The University of Tokyo
- **Jan–Feb 2024 · Visiting Researcher**  
University of Milano-Bicocca; host: Sebastiano Cantalupo

## Education

- **2017–2022 · Ph.D., SOKENDAI**  
Astronomy research based at NAOJ. Thesis: *Correlation between galaxy and IGM H I at  $z \approx 2$  mapped by Subaru/HSC*. Supervisors: Nobunari Kashikawa and Masayuki Tanaka.
- **2013–2017 · Bachelor's degree in astronomy**  
University of Science and Technology of China. Supervisor: Junxian Wang.

## Research funding

- **2024–2027 · Principal Investigator · JSPS KAKENHI 24K17084**  
Early-Career Scientists. Elucidating the Trigger and Feedback Mechanisms of the Most Concentrated Quasars at Cosmic Noon. JPY 4,550,000 including indirect costs; JPY 3,500,000 direct.
- **2025–2030 · Co-Investigator · JSPS KAKENHI 25H00663**  
Scientific Research (A). Galaxy Overdense Regions in the Distant Universe Revealed by Wide-Area Sky Survey and TAO. PI: Nobunari Kashikawa.

## Professional service & membership

- JCMT Time Allocation Committee, September 2023–November 2025.
- International Astronomical Union member, since June 2024.
- Astronomical Society of Japan member, since October 2017.

## Papers & preprints

2026

Jul 2026

Preprint

### Subaru meets JWST: A Direct Measurement of Ly $\alpha$ Escape Fraction at $z \approx 6.2$ with Dual Narrow-Band Imaging

Shimizu, Shunta et al.

*arXiv:2607.08264*

[ADS](#) [DOI](#) [arXiv](#)

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Jul 2026

### JWST COSMOS-3D: Spectroscopic census and luminosity function of [O III] emitters at $6.75 < z < 9.05$ in COSMOS

Meyer, Romain A. et al.

*A&A, 712, A16*

[ADS](#) [DOI](#) [arXiv](#)

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Jun 2026

Preprint

### Prime Focus Spectrograph on the Subaru Telescope: Overview of Science Operations

Tanaka, Masayuki et al.

*arXiv:2606.14012*

[ADS](#) [DOI](#) [arXiv](#)

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Jun 2026

### Subaru High- $z$ Exploration of Low-Luminosity Quasars (SHELLQs) — XXV. Large-scale environments of low-luminosity quasars at $z \sim 6$ traced by Ly $\alpha$ emitters

Arita, Junya et al.

*MNRAS, 549, stag784*

[ADS](#) [DOI](#) [arXiv](#)

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Apr 2026

### Cosmic Himalayas in CROCODILE: Probing the Extreme Quasar Overdensities by Count-in-cells Analysis and Nearest-neighbor Distribution

Kuwayama, Yuto et al.

*ApJ, 1001, 172*

[ADS](#) [DOI](#) [arXiv](#)

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Mar 2026 [Preprint](#)

## A Python-Based Peeling Framework for Radio Interferometry: Application to uGMRT 650MHz Imaging

Peng, Hao et al.

*arXiv:2603.10758*

[ADS](#) [DOI](#) [arXiv](#)

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Feb 2026

## Deeper detection limits in astronomical imaging using self-supervised spatiotemporal denoising

Guo, Yuduo et al.

*Science, 392, eady9404*

[ADS](#) [DOI](#) [arXiv](#)

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## 2025

Dec 2025 [Preprint](#)

## Little red dot variability over a century reveals black hole envelope via a giant Einstein cross

Zhang, Zijian et al.

*arXiv:2512.05180*

[ADS](#) [DOI](#) [arXiv](#)

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Jul 2025

## RIDEN pilot survey: broad-band selection of candidate quasars with extended Lyman- $\alpha$ nebulae using CLAUDS–HSC–SSP–DUNES<sup>2</sup> joint data

Shimakawa, Rhythm et al.

*MNRAS, 540, 3565–3582*

[ADS](#) [DOI](#) [arXiv](#)

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Jun 2025

## Cosmic Himalayas: The Highest Quasar Density Peak Identified in a 10,000 deg<sup>2</sup> Sky with Spatial Discrepancies between Galaxies, Quasars, and IGM H I

Liang, Yongming et al.

*ApJ, 986, 60*

[ADS](#) [DOI](#) [arXiv](#)

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Apr 2025

**SILVERRUSH. XIV. Ly $\alpha$  Luminosity Functions and Angular Correlation Functions from  $\sim 20,000$  Ly $\alpha$  Emitters at  $z \sim 2.2\text{--}7.3$  from up to  $24\text{ deg}^2$  HSC-SSP and CHORUS Surveys: Linking the Postreionization Epoch to the Heart of Reionization**

Umeda, Hiroya et al.

*ApJS*, 277, 37

[ADS](#) [DOI](#) [arXiv](#)

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Mar 2025

**The Physical Origin of Extreme Emission Line Galaxies at High Redshifts: Strong [O III] Emission Lines Produced by Obscured Active Galactic Nuclei**

Zhu, Chenghao et al.

*ApJ*, 982, 27

[ADS](#) [DOI](#) [arXiv](#)

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Mar 2025

**MAMMOTH-Subaru. IV. Large Scale Structure and Clustering Analysis of Ly $\alpha$  Emitters and Ly $\alpha$  Blobs at  $z = 2.2\text{--}2.3$**

Zhang, Haibin et al.

*ApJ*, 981, 70

[ADS](#) [DOI](#) [arXiv](#)

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Feb 2025

**Predicting Ly $\alpha$  emission from distant galaxies with neural network architecture**

Yoshioka, Takehiro et al.

*MNRAS*, 536, 3386–3400

[ADS](#) [DOI](#) [arXiv](#)

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## 2024

Dec 2024

**MAMMOTH-Subaru. II. Diverse Populations of Circumgalactic Ly $\alpha$  Nebulae at Cosmic Noon**

Li, Mingyu et al.

*ApJS*, 275, 27

[ADS](#) [DOI](#) [arXiv](#)

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Dec 2024

### Mining for Protoclusters at $z \sim 4$ from Photometric Data Sets with Deep Learning

Takeda, Yoshihiro et al.

*ApJ*, 977, 81

[ADS](#) [DOI](#) [arXiv](#)

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Sep 2024

### Ly $\alpha$ Imaging around the Hyperluminous Dust-obscured Quasar W2246–0526 at $z = 4.6$

Luo, Yibin et al.

*ApJ*, 972, 51

[ADS](#) [DOI](#) [arXiv](#)

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Jan 2024

### MAMMOTH-Subaru. V. Effects of Cosmic Variance on Ly $\alpha$ Luminosity Functions at $z = 2.2\text{--}2.3$

Ma, Ke et al.

*ApJ*, 961, 102

[ADS](#) [DOI](#) [arXiv](#)

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Jan 2024

### MAMMOTH-Subaru. III. Ly $\alpha$ Halo Identified by Stacking $\sim 3300$ Ly $\alpha$ Emitters at $z = 2.2\text{--}2.3$

Zhang, Haibin et al.

*ApJ*, 961, 63

[ADS](#) [DOI](#) [arXiv](#)

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## 2023

Sep 2023

### SILVERRUSH. XIII. A Catalog of 20,567 Ly $\alpha$ Emitters at $z = 2\text{--}7$ Identified in the Full-depth Data of the Subaru/HSC-SSP and CHORUS Surveys

Kikuta, Satoshi et al.

*ApJS*, 268, 24

[ADS](#) [DOI](#) [arXiv](#)

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Sep 2023

### Subaru High- $z$ Exploration of Low-luminosity Quasars (SHELLQs). XVIII. The Dark Matter Halo Mass of Quasars at $z \sim 6$

Arita, Junya et al.

*ApJ*, 954, 210

[ADS](#) [DOI](#) [arXiv](#)

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Mar 2023

### **COSMOS2020: Discovery of a Protocluster of Massive Quiescent Galaxies at $z = 2.77$**

Ito, Kei et al.

*ApJL*, 945, L9

[ADS](#)   [DOI](#)   [arXiv](#)

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## **2022**

Oct 2022

### **The physical origin for spatially large scatter of IGM opacity at the end of reionization: The IGM Ly $\alpha$ opacity-galaxy density relation**

Ishimoto, Rikako et al.

*MNRAS*, 515, 5914–5926

[ADS](#)   [DOI](#)   [arXiv](#)

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Apr 2022

### **Third data release of the Hyper Suprime-Cam Subaru Strategic Program**

Aihara, Hiroaki et al.

*PASJ*, 74, 247–272

[ADS](#)   [DOI](#)   [arXiv](#)

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Mar 2022

### **CHORUS. IV. Mapping the Spatially Inhomogeneous Cosmic Reionization with Subaru HSC**

Yoshioka, Takehiro et al.

*ApJ*, 927, 32

[ADS](#)   [DOI](#)   [arXiv](#)

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Feb 2022

### **A Wide and Deep Exploration of Radio Galaxies with Subaru HSC (WERGS). VI. Distant Filamentary Structures Pointed Out by High- $z$ Radio Galaxies at $z \sim 4$**

Uchiyama, Hisakazu et al.

*ApJ*, 926, 76

[ADS](#)   [DOI](#)   [arXiv](#)

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## 2021

Jul 2021

### Interrelation of the Environment of Ly $\alpha$ Emitters and Massive Galaxies at $2 < z < 4.5$

Ito, Kei et al.

*ApJ*, 916, 35

[ADS](#) [DOI](#) [arXiv](#)

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Jun 2021

### Probing Feedback via IGM tomography and the Ly $\alpha$ Forest with Subaru PFS, TMT/ELT, and JWST

Nagamine, Kentaro et al.

*ApJ*, 914, 66

[ADS](#) [DOI](#) [arXiv](#)

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Apr 2021

### SILVERRUSH X: Machine Learning-aided Selection of 9318 LAEs at $z = 2.2, 3.3, 4.9, 5.7, 6.6$ , and 7.0 from the HSC SSP and CHORUS Survey Data

Ono, Yoshiaki et al.

*ApJ*, 911, 78

[ADS](#) [DOI](#) [arXiv](#)

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Jan 2021

### Evidence for GN-z11 as a luminous galaxy at redshift 10.957

Jiang, Linhua et al.

*Nature Astronomy*, 5, 256–261

[ADS](#) [DOI](#) [arXiv](#)

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Jan 2021

### Statistical Correlation between the Distribution of Ly $\alpha$ Emitters and Intergalactic Medium H I at $z \sim 2.2$ Mapped by the Subaru/Hyper Suprime-Cam

Liang, Yongming et al.

*ApJ*, 907, 3

[ADS](#) [DOI](#) [arXiv](#)

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## 2020

Dec 2020

### **CHORUS. I. Cosmic HydrOgen Reionization Unveiled with Subaru: Overview**

Inoue, Akio K. et al.

*PASJ*, 72, 101

[ADS](#) [DOI](#) [arXiv](#)

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Dec 2020

### **Faint Quasars Live in the Same Number Density Environments as Lyman Break Galaxies at $z \sim 4$**

Uchiyama, Hisakazu et al.

*ApJ*, 905, 125

[ADS](#) [DOI](#) [arXiv](#)

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Nov 2020

### **Subaru High- $z$ Exploration of Low-luminosity Quasars (SHELLQs). XI. Proximity Zone Analysis for Faint Quasar Spectra at $z \sim 6$**

Ishimoto, Rikako et al.

*ApJ*, 903, 60

[ADS](#) [DOI](#) [arXiv](#)

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Aug 2020

### **The UV Luminosity Function of Protocluster Galaxies at $z \sim 4$ : The Bright-end Excess and the Enhanced Star Formation Rate Density**

Ito, Kei et al.

*ApJ*, 899, 5

[ADS](#) [DOI](#) [arXiv](#)

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## 2019

Dec 2019

### **Second data release of the Hyper Suprime-Cam Subaru Strategic Program**

Aihara, Hiroaki et al.

*PASJ*, 71, 114

[ADS](#) [DOI](#) [arXiv](#)

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Jun 2019

### **The Brightest UV-selected Galaxies in Protoclusters at $z \sim 4$ : Ancestors of Brightest Cluster Galaxies?**

Ito, Kei et al.

*ApJ*, 878, 68

[ADS](#)   [DOI](#)   [arXiv](#)

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Jan 2019

### **Suppression of Low-mass Galaxy Formation around Quasars at $z \sim 2-3$**

Uchiyama, Hisakazu et al.

*ApJ*, 870, 45

[ADS](#)   [DOI](#)   [arXiv](#)

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